ViGoEmotions: Vietnamese Emotion Recognition

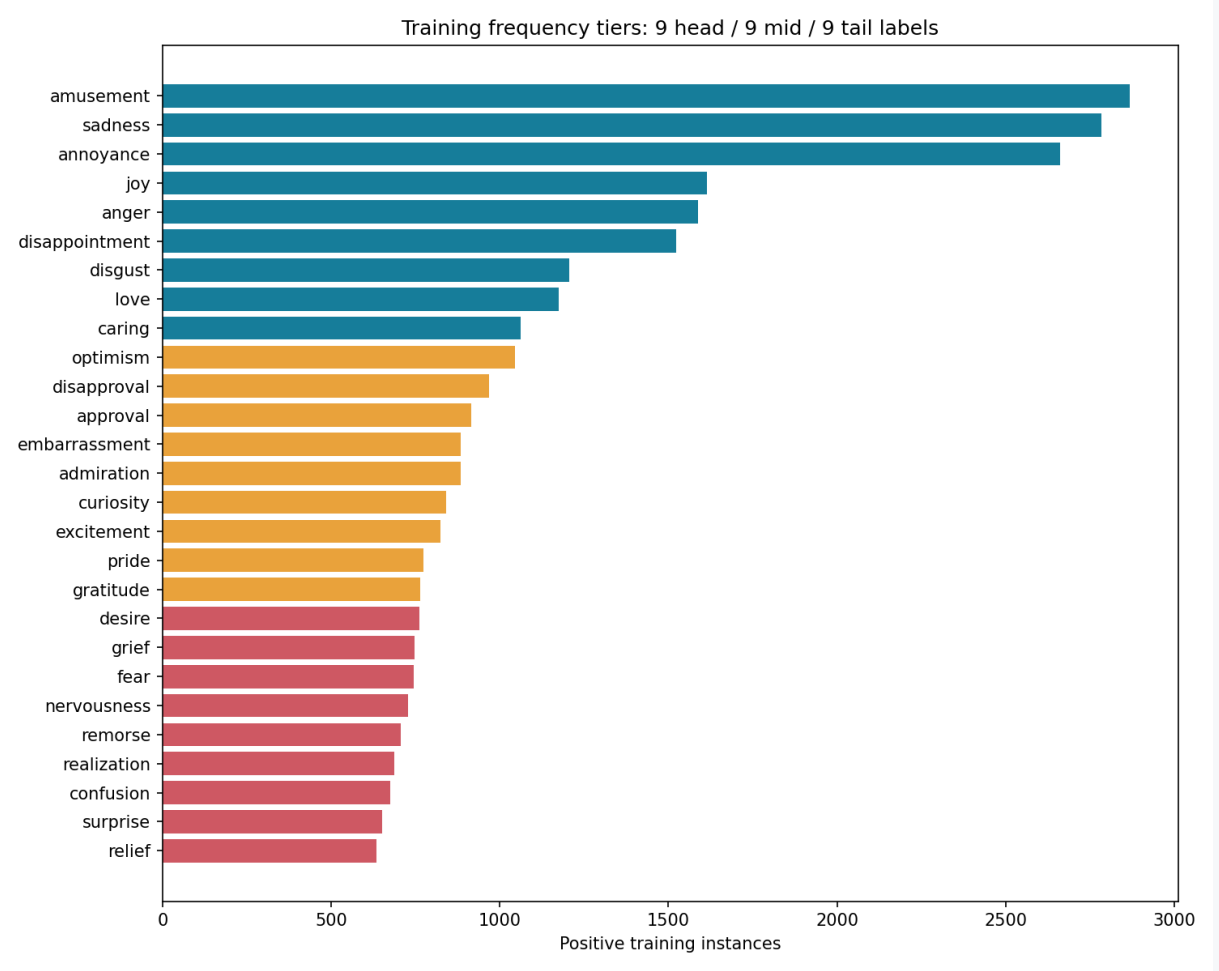
- Multi-label classification of Vietnamese comments.
- Three models; 27 emotions or 28 labels with Neutral.
- Both neural models: 20 epochs in each setting.
- All reported model scores are validation results.

Dataset and Quality Checks

- 20,664 comments; supplied train/validation/test split.
- One train–test text overlap removed from training.
- Effective sizes: 16,530 / 2,066 / 2,067.
- Source annotations: 725 Neutral-only; 306 mixed.

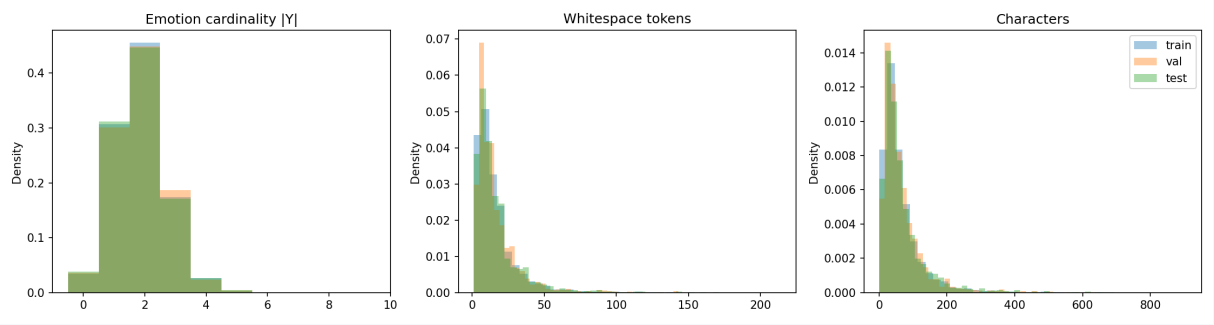
Emotion Frequency and Imbalance

- 27 emotions; nine labels per frequency tier.
- Amusement is most frequent; relief is least frequent.
- Maximum/minimum training count: about 4.52×.
- Positive labels are sparse across the output vector.



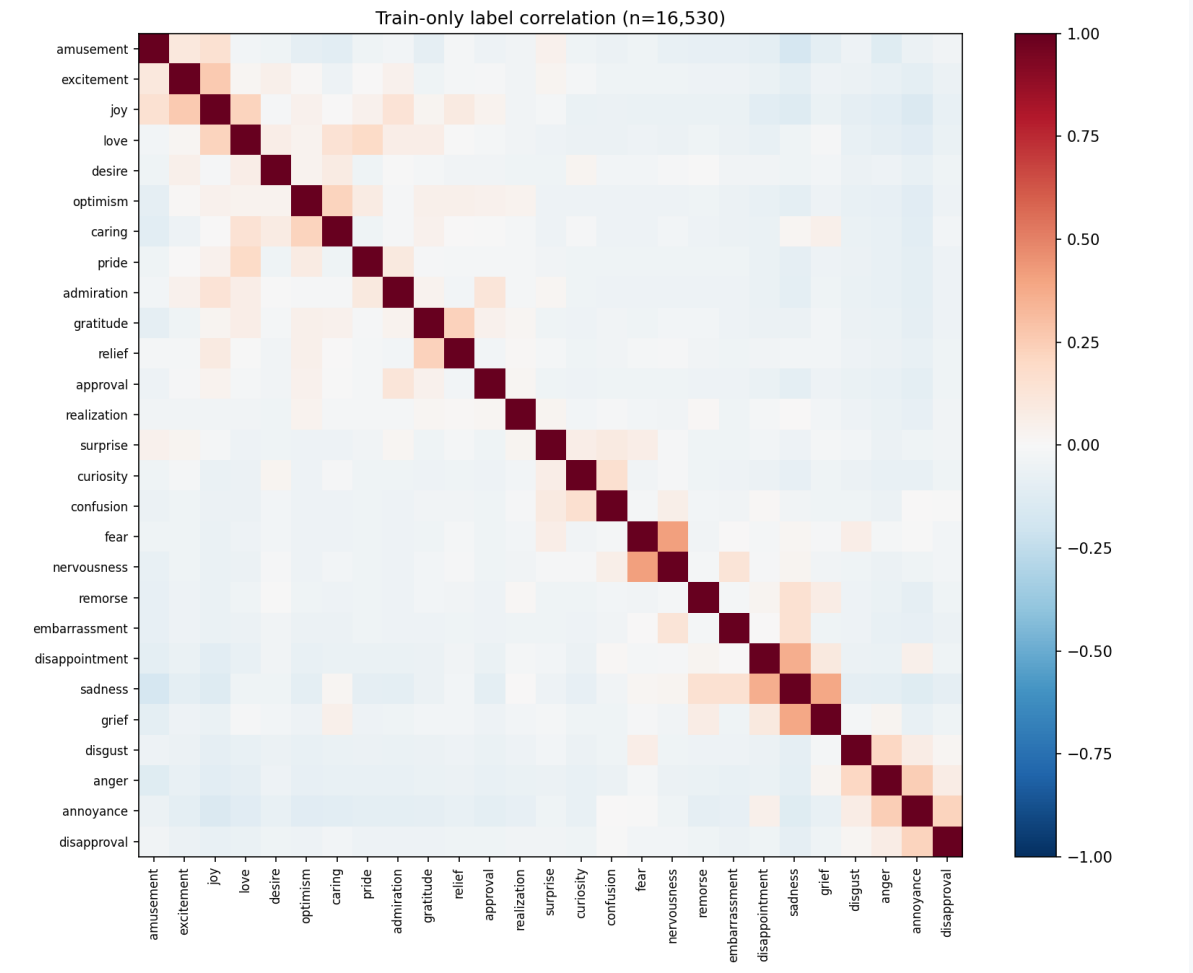
Label Cardinality and Text Length

- Training mean: 1.859 emotions per comment.
- Maximum: five emotions in one comment.
- Median: 11 whitespace tokens; 47 characters.
- Informal language: slang, teencode, emoji and HTML.



Emotions Often Occur Together

- Disappointment–sadness: 904 training comments.
- Anger–annoyance: 699; sadness–grief: 620.
- Co-occurrence and correlation use training labels.
- Correlation does not establish a causal relationship.



Split Protocol and Leakage Prevention

- Preserve the supplied split and fixed sample IDs.
- Fit vocabulary and IDF on training data only.
- Normalize Unicode and whitespace; preserve negation.
- Distribution audit does not prove stratification.
- No model evaluation or threshold tuning on test.

Two Label Settings: 27 vs 28

- 27 outputs: Neutral-only becomes an all-zero vector.
- 28 outputs: Neutral is a separate learned label.
- Preserve mixed annotations; no exclusion rule.
- Validation: 75 Neutral-only + 19 mixed = 94.
- Independent sigmoid outputs with binary cross-entropy.

Three Models, One Experimental Profile

- TF-IDF + OvR LR: liblinear, max_iter = 1,000.
- BiGRU: embedding/hidden 128; batch 64; lr 0.001.
- PhoBERT: full fine-tuning; batch 16; lr 2e-5.
- BiGRU and PhoBERT: 20 epochs for both label sets.
- TF-IDF uses a solver limit, not neural epochs.

Checkpoint Selection and Reproducibility

- Seed 42; threshold 0.5; same split and settings.
- Select by validation Macro-F1 on the 27 emotions.
- Best BiGRU epochs, 27/28: 17 / 20.
- Best PhoBERT epochs, 27/28: 14 / 15.
- Same GPU model and recorded package versions.

Results: 27 Emotions

- TF-IDF Macro-F1: 0.2388.
- BiGRU Macro-F1: 0.4220.
- PhoBERT Macro-F1: 0.5974; Micro-F1: 0.5996.
- PhoBERT Hamming Loss: 0.0550; LRAP: 0.7419.
- Higher F1/LRAP is better; lower Hamming Loss is better.

Results: 28 Labels Including Neutral

- TF-IDF Macro-F1: 0.2310.
- BiGRU Macro-F1: 0.4086.
- PhoBERT Macro-F1: 0.5783; Micro-F1: 0.5829.
- PhoBERT Hamming Loss: 0.0575; LRAP: 0.7077.
- These metrics include the additional Neutral label.

Does Adding Neutral Help the 27 Emotions?

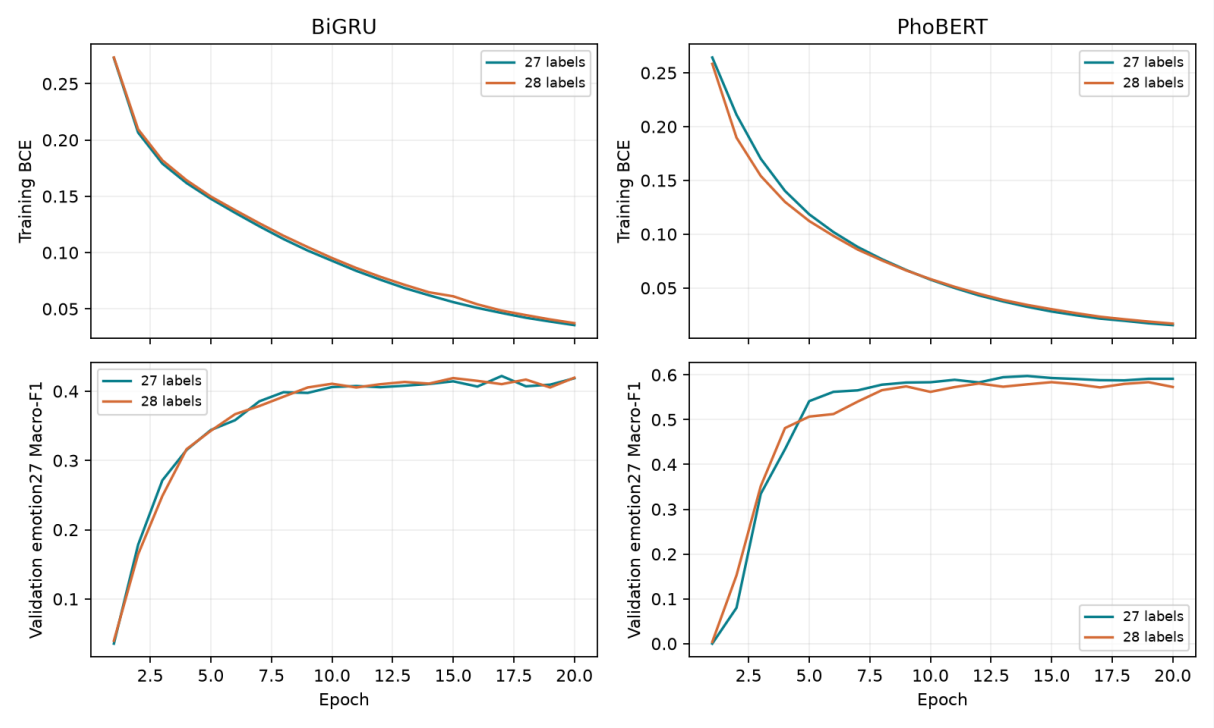
- PhoBERT with 27 outputs: 0.5974.
- PhoBERT with 28 outputs, scored on 27: 0.5836.
- Difference: -1.38 percentage points.
- No observed improvement on the shared 27 emotions.
- One seed; no causal or statistical conclusion.

Neutral Detection Remains Difficult

- PhoBERT Neutral F1: 0.4369.
- Precision: 0.4018; recall: 0.4787.
- 45 TP / 67 FP / 49 FN; support: 94.
- No Neutral output exists in the 27-label model.
- An empty prediction is not automatically Neutral.

Learning Curves Across 20 Epochs

- All four neural runs completed epochs 1–20.
- The best checkpoint can occur before the last epoch.
- PhoBERT 27 starts at F1 = 0, then improves.
- Training loss alone cannot confirm overfitting.



Q1–Q2: Per-Label F1 and Frequency

- PhoBERT best label: gratitude, F1 0.8370.
- Lowest label: disapproval, F1 0.3623.
- Frequency vs F1: Spearman rho 0.018.
- p-value 0.928; descriptive evidence only.
- Full per-label results are included in the report.

Q3–Q4: Label Pairs and Text Styles

- Disappointment–sadness: PhoBERT gets both in 63/122 cases.
- Joint recall at 0.5: 51.64%.
- Largest PhoBERT–BiGRU style gap: teencode.
- Macro-F1 gap in this group: +22.13 percentage points.
- Style groups use heuristics and may overlap.

Q5: Sensitivity to the Decision Threshold

- PhoBERT 27 at 0.5: Macro-F1 0.5974.
- Best validation grid point: 0.35, F1 0.5978.
- Same checkpoint; exploratory validation analysis.
- Very small F1 gain; Hamming Loss increases.
- Main results remain at threshold 0.5.

Error Analysis: New Neutral Predictions

- Six PhoBERT 28 cases: three Neutral FP and three FN.
- Inspect implied emotion, short text, HTML and profanity.
- Publish IDs and abstract observations, not raw comments.
- Preliminary AI-assisted review; team confirmation pending.
- Error candidates are not a completed manual audit.

Final Research Question and Next Steps

- Is one threshold suitable for every emotion?
- Freeze the best neural checkpoint; keep weights fixed.
- Compare 0.5, global tuned and per-label thresholds.
- Tune on validation; lock choices; evaluate test once.
- Review at least 20 false negatives and 20 false positives.

Conclusions and Evidence

- PhoBERT leads within both label settings.
- Adding Neutral does not improve shared-emotion F1 in this run.
- Evidence: notebooks 08/09, saved predictions and manifests.
- Limits: one seed, validation selection, no test scores.
- Sources: course brief, ViGoEmotions, PhoBERT and libraries.